EXP NO: 8 (version 2)

MODEL EVALUATION AND IMPROVEMENT: HYPERPARAMETER TUNING WITH GRID SEARCH AND CROSS-VALIDATION

AIM

To evaluate and improve a machine learning model using **Grid Search for hyperparameter tuning** and **K-Fold Cross-Validation** to obtain reliable model performance and reduce overfitting.

PROCEDURE

1. Load the Iris dataset.
2. Divide the data into training and testing sets.
3. Normalize the feature values.
4. Define different SVM hyperparameters.
5. Perform Grid Search using 5-fold validation.
6. Test the optimized SVM model.
7. Perform K-Fold Cross-Validation.
8. Display the accuracy results graphically.

CODE

```
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split, GridSearchCV, KFold, cross_val_score
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix, ConfusionMatrixDisplay

data = load_iris()

X = data.data
y = data.target

X_train, X_test, y_train, y_test = train_test_split(
 X, y, test_size=0.25, random_state=7, stratify=y
)

pipeline = Pipeline([
 ("scaler", StandardScaler()),
 ("svm", SVC())
])

parameters = {
 "svm__C": [0.5, 1, 5, 20],
```

```
 "svm__gamma": ["scale", 0.1, 0.01],
"svm__kernel": ["rbf", "linear"]
 }

search= GridSearchCV(
 pipeline,
 parameters,
 cv=5,
 scoring="accuracy"
)

search.fit(X_train, y_train)

print("Best Parameters:")
print(search.best_params_)

print("Best Cross-Validation Accuracy:")
print(round(search.best_score_, 4))

best_model= search.best_estimator_

predictions= best_model.predict(X_test)

accuracy= accuracy_score(y_test, predictions)

print("Test Accuracy:", round(accuracy, 4))

matrix= confusion_matrix(y_test, predictions)

ConfusionMatrixDisplay(
 confusion_matrix=matrix,
 display_labels=data.target_names
).plot()

plt.title("Confusion Matrix - Optimized SVM")
plt.show()

kf= KFold(
 n_splits=5,
 shuffle=True,
 random_state=7
)

scores= cross_val_score(
 pipeline,
 X,
 y,
 cv=kf,
 scoring="accuracy"
)

print("Cross-Validation Scores:")
print(np.round(scores, 4))
```

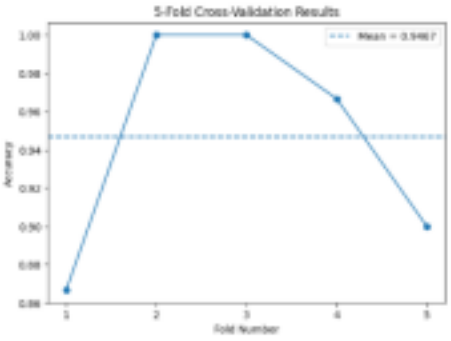
```
mean_score= scores.mean()
std_score= scores.std()

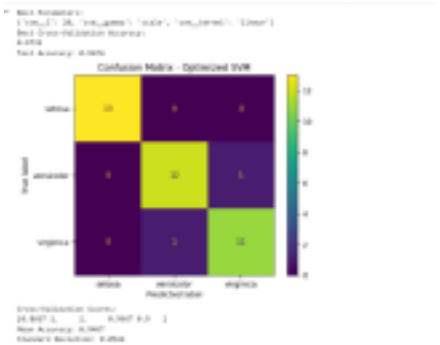


print("Mean Accuracy:", round(mean_score, 4))
print("Standard Deviation:", round(std_score, 4))

plt.figure(figsize=(7, 5))
plt.plot(
 range(1, 6),
 scores,
 marker="o"
)
plt.axhline(
 mean_score,
 linestyle="--",
 label=f"Mean = {mean_score:.4f}"
)
plt.xticks(range(1, 6))
plt.xlabel("Fold Number")
plt.ylabel("Accuracy")
plt.title("5-Fold Cross-Validation Results")
plt.legend()
plt.show()
```

OUTPUT:





RESULT

The SVM classifier was successfully optimized using **Grid Search** and evaluated using **5-Fold Cross-Validation**. The best hyperparameter combination was obtained automatically, and the model showed good classification accuracy on the Iris dataset